Thrift

A coding agent that spends fewer tokens on the fix.

EARLY BENCHMARK

86%

fewer tokens per run on four shared simple tasks

Small bugs can carry a big context bill.

Every turn repeats work.

A baseline lists files, reads them one by one and resends the full history for every edit and test command.

Old API habits survive.

An Express 4 wildcard can crash an Express 5 service before the first request reaches it.

Route once. Keep the prompt short.

01 / CONTEXT

Find the code

Rank files against the task. Put the useful files in the first coding prompt.

02 / BIFROST

Choose a model

Classify the task once, then pin the model that answers.

03 / TOOLS

Apply the edit

Run workspace tools through MCP. Ask before a write or shell command.

04 / TESTS

Stop on green

Run the tests after an edit. Trim old tool output if another step is needed.

Thrift sends step events to the dashboard. Bifrost logs supply routing, fallback and cost data.

Gateway controls in each run.

Routing	Local embeddings classify the task. CEL rules route simple tasks to GPT-OSS 20B and harder tasks to GPT-OSS 120B.
Fallbacks	The configured chain starts at Groq, moves to Gemini and can finish on local Ollama.
Virtual keys	Separate keys control baseline access, daily spend and local-only inference.
MCP and cache	Thrift executes approved tool calls through Bifrost. Redis backs the gateway cache.

Cost comes from the gateway logs, including the linked fallback requests.

Check the installed version.

From claim to source

Thrift reads installed package versions, asks a model which API claims the fix depends on and searches Tavily for evidence.

The judge attaches a source and puts outdated or uncertain claims into the coding context.

Express 5 example

Wildcard routes require a named parameter. The Express 4 pattern throws when the server registers it.

```
/api/*  →  /api/*rest
```

Stale mode turns off this check. Both model calls count toward thrift's tokens.

11,376 to 1,621 tokens.

Mode	Valid runs	Pass rate	Avg tokens	Avg model cost
Naive	11	100%	11,376	$0.00159
Thrift	12	100%	1,621	$0.00023

Shared tasks 01-04 are all simple. One thrift cost is missing. In a separate fallback batch, thrift passed 10 of 11 tasks in the follow-up batch. Groq returned HTTP 403, so those tasks have no naive comparison.

Source: `bench/results/run-2026-11-07T06-17-12-387Z.jsonl`

Both modes passed 2 of 9.

Mode	Passing runs	Recorded avg tokens
Stale	2 / 9	14,789
Thrift	2 / 9	11,062

The check caught the old Zod accessor twice. Express verdicts contained errors, and two checks stopped early. Coding models varied across attempts, so the token gap cannot be attributed to Tavily.

Source: `bench/results/run-2026-11-09T12-37-18-207Z.jsonl`